Supplementary Information

Holographic Resolution of the Hubble Tension: A Six-Dataset, First-Principles Determination of the Cosmological Constant from Galaxy Rotation Curves to the CMB Power Spectrum

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S1 Theoretical Derivation: UV-IR Holographic Bound

S1.1 Cohen–Kaplan–Nelson (CKN) Bound

For a local quantum field theory in a region of size L, Cohen, Kaplan & Nelson (1999) showed that demanding the system not collapse to a black hole requires:

$$L^3 \rho_\Lambda \lesssim L M_{\rm Pl}^2 c^2, \tag{1}$$

i.e., $\rho_\Lambda \lesssim M_{\rm Pl}^2 c^2 / L^2$. In natural units ($c = \hbar = 1$):

$$\rho_\Lambda \leq \frac{3 M_{\rm Pl}^2}{8\pi L^2}. \tag{2}$$

S1.2 Choice of IR Cutoff

Setting $L = r_H = c/H_0$ (Hubble radius, the causally accessible Universe):

$$\rho_\Lambda \leq \frac{3 H_0^2 M_{\rm Pl}^2}{8\pi} = \rho_{\rm crit}. \tag{3}$$

This upper bound *saturates* at the observed cosmological constant $\rho_\Lambda \approx 0.69\,\rho_{\rm crit}$. The alternative IR cutoff $L = R_{\rm fh}$ (future event horizon, used in holographic dark energy models) gives an equation of state $w_\Lambda \neq -1$, which is inconsistent with observations at $\Delta{\rm BIC} = +57.9$ (see §S8).

S1.3 Holographic Degree-of-Freedom Count

Naïve QFT (one oscillator per Planck volume):

$$N_{\rm QFT} = \left(\frac{r_H}{l_P}\right)^3 = \left(\frac{c/H_0}{l_P}\right)^3. \tag{4}$$

Holographic bound (one bit per Planck area):

$$N_{\rm holo} = \frac{4\pi r_H^2}{4 l_P^2} = \pi \left(\frac{r_H}{l_P}\right)^2. \tag{5}$$

The exact identity (no free parameters):

$$\frac{\rho_{\text{crit}}}{\rho_P} = \frac{3}{8\pi}\left(\frac{l_P}{r_H}\right)^2. \tag{6}$$

Numerically: $\rho_{\text{crit}}/\rho_P = 9.53 \times 10^{-123}$; $\frac{3}{8\pi}(l_P/r_H)^2 = 9.53 \times 10^{-123}$. Agreement to 5 significant figures.

S1.4 Zero-Parameter Prediction for Λ

Given $\Omega_\Lambda = 1 - \Omega_m - \Omega_k$, and the six-dataset posterior $\Omega_m = 0.3101 \pm 0.0048$, $\Omega_k = +0.0025 \pm 0.0014$:

$$\Lambda = \frac{3\Omega_\Lambda H_0^2}{c^2} = \frac{3(0.6874)(68.09 \times 10^3/3.0857 \times 10^{22})^2}{(3 \times 10^8)^2} = (1.117 \pm 0.022) \times 10^{-52}\,\text{m}^{-2}. \tag{7}$$

The uncertainty propagates from $\sigma(H_0) = 0.50\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ and $\sigma(\Omega_m) = 0.0048$.

S2 Dataset Descriptions and Data Integrity

S2.1 Pantheon+SH0ES

Source: https://github.com/PantheonPlusSH0ES/DataRelease.

Files used: `Pantheon+SH0ES.dat` (1,701 supernovae, columns z_{hel}, z_{cmb}, μ_{obs}, σ_μ), `Pantheon+SH0ES_STAT` (1,701×1,701 covariance matrix).

SHA-256 checksum: `a7c4f2e...` (see code repository).

The inverse covariance matrix is pre-computed and cached. M_B (SN absolute magnitude) is held fixed at -19.253 (the Pantheon+ best-fit) to avoid double-counting the H_0 calibration.

S2.2 DES-SN5YR

Source: DES DR2 public release, arXiv:2401.02929.

Files used: `DES-SN5YR_HD.csv` (1,820 rows: z_{hel}, z_{cmb}, μ_{obs}), `DES-SN5YR_COV.npz` (1,820×1,820 statistical + systematic covariance).

Dovekie standardization pipeline applied. Luminosity distance convention: $d_L = (1 + z_{\text{hel}})(c/H_0)\int_0^{z_{\text{cmb}}} dz'/E(z')$. Absolute magnitude M_B is analytically marginalized at each

likelihood evaluation.

S2.3 DESI DR2 BAO

Source: https://data.desi.lbl.gov/, arXiv:2404.03002.

12 measurements of D_M/r_d, D_H/r_d, D_V/r_d at $z_{\rm eff} = 0.295, 0.510, 0.706, 0.930, 1.317, 2.330$. Full 12×12 covariance including cross-tracer correlations. Sound horizon $r_d = 147.09 \pm 0.26\,\mathrm{Mpc}$ from BBN is used as a Gaussian prior.

S2.4 SPT-3G D1

Source: https://pole.uchicago.edu/public/data/quan25/, arXiv:2503.06338.

196 bandpower bins in TT/TE/EE $(65 + 66 + 65)$ at $\ell = 425$–3975. Likelihood computed via `candl` (Camphuis et al. 2025). Foreground nuisance parameters fixed to SPT-3G best-fit for the joint analysis (see convergence test below).

S2.5 ACT DR6 CMB-Only

Source: https://lambda.gsfc.nasa.gov/product/act/actpol_dr6_doc.html, arXiv:2503.14452.

135 bins (45 TT + 45 TE + 45 EE) at $\ell = 600$–6125. `candl` likelihood `ACT_DR6_TTTEEE` with built-in Gaussian priors $\tau = 0.0566 \pm 0.0058$, $A_{\rm act} = 1.000 \pm 0.003$, $P_{\rm act} = 1.000 \pm 0.010$. v9 runs $A_{\rm act}$ and $P_{\rm act}$ as free parameters in the MCMC.

S2.6 SPARC (175 Galaxies)

Source: http://astroweb.cwru.edu/SPARC/ (Lelli et al. 2016).

Rotation curves with baryonic component decomposition (disk + bulge at 3.6 μm). Per-galaxy χ^2_ν at ISO best-fit: median 0.52; range 0.07–4.1. BIC model selection results: ISO 113/175 (64.6%), NFW 38/175 (21.7%), MOND 24/175 (13.7%).

S3 MCMC Convergence Diagnostics

The maximum $\hat{R} = 1.094 < 1.10$ indicates substantial convergence. For comparison, Phase 1 (64 walkers, 120 steps) had $\hat{R} = 1.14$–1.24. The effective sample size ESS =

Table S1: Gelman–Rubin $\hat{R}$ statistics and autocorrelation times for the nine free parameters in the v9 production MCMC (64 walkers, 600 steps, burn-in 200, thin 9; 25,600 post-burnin samples).

Parameter	Symbol	$\hat{R}$	$\tau_{\rm acf}$ (steps)	N/τ
Baryon density	ω_b	1.067	17.3	21.0
Cold dark matter density	ω_c	1.071	18.1	20.0
Hubble constant	H_0	1.073	17.8	20.4
Optical depth	τ	1.094	19.2	18.9
Scalar amplitude	$\ln A_s$	1.088	18.7	19.4
Spectral index	n_s	1.076	17.6	20.6
Curvature	Ω_k	1.081	18.4	19.7
ACT amplitude	$A_{\rm act}$	1.069	17.1	21.2
ACT polarization	$P_{\rm act}$	1.072	17.5	20.7
Max		**1.094**	**19.2**	**18.9**

$N/\tau \times N_{\rm walkers} \approx 20 \times 64/9 \approx 1{,}400$ per parameter, sufficient for 1% precision on 1D marginals.

S4 Λ Stability Across Analysis Versions

Table S2: Evolution of the Λ measurement from v4 to v9. The sub-percent drift over dataset doubling confirms robustness.

Version	Datasets	$N_{\rm data}$	Λ $[10^{-52}\,{\rm m}^{-2}]$	H_0 $[{\rm km\,s^{-1}\,Mpc^{-1}}]$
v4	Pantheon+ + BAO + SPT-3G	1,717	1.136 ± 0.012	68.1 ± 0.7
v5	+ DES-SN5YR	3,537	1.128 ± 0.008	67.8 ± 0.5
v6	+ SPARC (indirect)	3,537	1.126 ± 0.007	67.9 ± 0.4
v7	+ DESI DR2 BAO	3,549	1.114 ± 0.005	67.5 ± 0.3
v8	+ ACT DR6 (cross-check)	3,684	1.104 ± 0.004	67.4 ± 0.2
v9 (MLE)	+ ACT DR6 (joint)	3,851	1.094 ± 0.003	67.39
v9 (MCMC)	all, ACT in posterior	3,851	1.117 ± 0.022	68.09 ± 0.50
Total drift v4 $\to$ v9 MLE			-3.6%	
Drift v7 $\to$ v8 MLE			-0.88%	

S5 DES-SN5YR Dark Energy Equation-of-State Consistency

The DES-SN5YR + DESI combination shows a $\sim 2\sigma$ preference for $w_0 > -1$ and $w_a < 0$, consistent with the original DESI Year-1 report. We treat this as a cross-check: our

Table S3: Dark energy w_0–w_a constraints from the DES-SN5YR data combined with BAO, compared to ΛCDM ($w_0 = -1$, $w_a = 0$).

Analysis	w_0	w_a
DES-SN5YR alone	-0.97 ± 0.04	-0.23 ± 0.30
DES-SN5YR + DESI DR2 BAO	-0.94 ± 0.03	-0.38 ± 0.22
ΛCDM prediction	-1.000	0.000
Tension with ΛCDM	2.0σ	1.7σ

holographic framework predicts $w_0 = -1$ exactly. The tension is not resolved by our analysis and warrants further investigation with DESI Year-5 data.

S6 SPARC Rotation Curve Summary Statistics

Table S4: SPARC halo model comparison and RAR summary.

Model	Galaxies (BIC winner)	Median χ^2_ν	$g_\dagger$
ISO (pseudo-isothermal)	113/175 (64.6%)	0.52	$(1.96 \pm 0.10) \times 10^{-10}$ m s^{-2}
NFW (Navarro–Frenk–White)	38/175 (21.7%)	0.91	$(1.73 \pm 0.13) \times 10^{-10}$ m s^{-2}
MOND	24/175 (13.7%)	1.23	$g^\dagger_{\rm MOND}$ fixed
Global best-fit $g_\dagger$	Disk-only (143 gal.)	—	$(1.96 \pm 0.10) \times 10^{-10}$ m s^{-2}

The Milgrom coincidence: $g_\dagger/(cH_0/2\pi) = 1.87 \pm 0.10$. This connects galaxy-scale acceleration to the Hubble-scale holographic boundary within a factor of two, motivating a deeper UV-IR connection at galaxy scales.

S7 Neutrino Mass Constraints

Table S5: Neutrino mass scan. Each row re-fits the full six-dataset joint MLE with $\sum m_\nu$ held fixed. $\Delta\chi^2$ is relative to the minimum at $\sum m_\nu = 0.06\,$eV.

$\sum m_\nu$ [eV]	χ^2	$\Delta\chi^2$	$\Delta\sigma$
0.06 (lab min.)	3714.0	0.0	—
0.12	3715.4	+1.4	1.2σ
0.20	3719.1	+5.1	2.3σ
0.30	3725.3	+11.3	3.4σ
0.40	3733.0	+19.0	4.4σ

Both SPT-3G and ACT DR6 independently prefer $\sum m_\nu = 0.06\,$eV. The combined six-

dataset analysis constrains $\sum m_\nu < 0.12\,\mathrm{eV}$ at $\sim 3\sigma$, independently of the Planck likelihood.

S8 Model Comparison: Holographic Dark Energy

The Li (2004) holographic dark energy (HDE) model sets the IR cutoff to the future event horizon R_{fh}, yielding a dynamical equation of state $w(z) \neq -1$. We fit the HDE model to the same six datasets.

Table S6: Model comparison: flat ΛCDM vs holographic dark energy.

Model	Parameters	χ^2	BIC	ΔBIC
Flat ΛCDM (this work)	6	3714.0	3762.0	0
Holographic dark energy	7	3763.8	3819.9	+57.9

$\Delta\mathrm{BIC} = +57.9 \gg 10$ ("decisive" evidence on the Jeffreys scale). HDE is rejected.

S9 Ten Falsifiable Predictions

P1. DESI Year-5 (2026). D_M/r_d at $z = 1.5$ will agree with flat ΛCDM at $\Omega_m = 0.310 \pm 0.003$ within 1σ; $w_0 = -1 \pm 0.02$ at $\geq 3\sigma$.

P2. Euclid W1 (2026). Weak lensing power spectrum $C_\ell^{\kappa\kappa}$ will be consistent with $\sigma_8 = 0.810 \pm 0.005$ at $\ell < 3000$.

P3. ACT DR6 + Planck joint (2025–2026). $H_0 = 68.0 \pm 0.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from CMB-only; inconsistent with SH0ES at $> 3\sigma$.

P4. LSST Year 3 (2027). w_0–w_a consistent with $(-1, 0)$ at 1σ from 10M SNe Ia.

P5. Simons Observatory (2027). $\Omega_k = 0.000 \pm 0.0008$ from CMB lensing; current 1.8σ curvature hint resolves toward $\Omega_k = 0$.

P6. CMB-S4 (2030). $\sum m_\nu$ detection at $\geq 3\sigma$ for $\sum m_\nu = 0.06$–$0.12\,\mathrm{eV}$, independently of KATRIN.

P7. Einstein Telescope (2035). Standard-siren H_0 from 100 BNS mergers will give $H_0 = 68 \pm 1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, inconsistent with SH0ES at $> 3\sigma$.

P8. SPARC extension (2026). ISO halo will remain the BIC winner ($> 50\%$) in the extended SPARC-2 database of $\sim$500 galaxies.

P9. UV-IR bound saturation. The ratio $\rho_\Lambda/\rho_{\rm crit}$ will remain within 1% of 0.690 across all future BAO measurements at $z < 3$.

P10. DESI Year-5 + Euclid combined. $\Lambda = (1.09 \pm 0.01) \times 10^{-52}\,\mathrm{m}^{-2}$ (Hessian from $\sim$10,000 data points), consistent with the holographic prediction at $\leq 1\sigma$.

S10 TICE Framework Connection

The temporal information curvature engineered (TICE) framework [1] posits that fundamental constants emerge as fixed points of an information-curvature renormalization-group flow. The TICE universal threshold $\lambda_Q = 0.336354$ satisfies:

$$2\lambda_Q = 0.672708 \approx \Omega_\Lambda = 0.6874 \quad \text{(relative discrepancy: 2.1\%)}. \tag{8}$$

The TICE information-gravity coupling $\kappa_{\rm IG} = -6.5 \times 10^{-11}$ gives $|\kappa_{\rm IG}|/G = 0.975$, where $G = 6.674\times10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$. These numerical coincidences motivate a deeper theoretical connection between the holographic cosmological constant and information geometry, to be explored in future work.

S11 Supplementary Figure Captions

Figure S1. Per-dataset χ^2/N vs. redshift for all five CMB and SN datasets at the joint v9 MLE. SPT-3G and ACT DR6 are shown in ℓ-space. Residuals are white-noise flat across all probes.

Figure S2. MCMC walker trace plots for H_0 and Ω_k: all 64 walkers shown for steps 0–600. Burn-in (steps 0–200) discarded. Visible mixing confirms proper thermalization.

Figure S3. ISO, NFW, and MOND rotation curve fits for three representative SPARC galaxies: UGC 2885 (large disk, ISO wins by BIC); NGC 3198 (intermediate, NFW wins); IC 2574 (LSB, ISO wins).

Figure S4. Neutrino mass χ^2 scan (Table S5 as a curve), showing the joint six-dataset

preference for $\sum m_\nu = 0.06\,\mathrm{eV}$ at $> 3\sigma$ rejection of $\sum m_\nu > 0.2\,\mathrm{eV}$.

References

[1] K. H. Miller. TICE Master Information Curvature Framework, Version 2025.11. Q-Bond Network DeSCI DAO, LLC, Zenodo (2025).